Akoizo Original Practice - Form 2 - Answer key

Division C | Ping Pong Parachute | 50-minute written practice

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

ANSWER KEY - Review only after submitting.

1. (2 points) Why is the powered phase so short in this event?

Unit: How an air-powered bottle rocket climbs.
Worked-case data: A rocket reaches apogee 1.2 s after launch, and the payload then descends for 11.5 s. What share of time aloft came from the descent?
Separate investigation observations:
40 psi: Peak about 12 ft; time aloft 8.1 s.
55 psi: Peak about 18 ft; time aloft 12.4 s.
70 psi: Peak about 24 ft; time aloft 16.0 s in a 30 ft hall.

A. Fins stop the thrust

B. The parachute opens immediately

C. Only air is used, so the propellant is very light and leaves almost instantly

D. The launcher limits it to one second

Correct: C

2. (2 points) Which set meets the minimum practice log requirement?

Unit: The practice log and launch decisions.
Worked-case data: A log has 12 flights recording pressure, height, time, and fin count. Is it complete?
Separate investigation observations:
Log entries: 14 flights recorded.
Columns: Pressure, peak height, time aloft, parachute diameter.
Flights 1-3: Peak height column left blank.

A. 10 flights with pressure, height, time, parachute diameter, and rocket mass

B. 20 flights with pressure, time, and parachute diameter

C. 9 flights with six parameters

D. 10 flights with pressure and time only

Correct: A

3. (2 points) At terminal velocity, the drag on the payload is:

Unit: Parachute physics: drag and slow descent.
Worked-case data: A 4.7 g payload uses a 40 cm diameter dome (C_d about 1.5). Estimate its descent speed and the time to fall 7 m.
Separate investigation observations:
30 cm canopy: Opens every time; 7.8 s descent.
60 cm canopy: Opens 6 of 10 times; 14.9 s when open, 3 s when not.
50 cm canopy, looser fold: Opens 10 of 10; 12.6 s descent.

A. Equal to its weight

B. Zero

C. Twice its weight

D. Less than its weight

Correct: A

4. (2 points) Which material is NOT allowed anywhere on the rocket or parachute?

A. Mylar film

B. Plastic trash bag

C. Foam board

D. Paper tape

Correct: A

5. (4 points) Why is the second launch valuable even after a good first launch?

answer: It can beat the first launch, and its time is the tiebreaker if teams tie on their best launch. (4 points).

6. (4 points) Why can’t a team use a spring to eject the parachute?

Unit: Deployment and separation.
Worked-case data: Video shows a payload leaving the rocket 0.3 s after launch, well before apogee, and the parachute tangling. What should change?
Separate investigation observations:
Cup depth 2 cm: Payload left during climb in 4 of 5 flights.
Cup depth 4 cm: Payload stayed attached in 3 of 5 flights.
Cup depth 3 cm: Payload released near apogee in 5 of 5 flights.

answer: All energy must come from launcher air pressure; springs and elastic assists are specifically prohibited. (4 points).

7. (4 points) Why should your launch pressure depend on the spread of your data, not only the average?

Unit: Apogee, ceiling height, and choosing pressure.
Worked-case data: Your chart shows 50 psi -> 17 ft, 60 psi -> 21 ft, 70 psi -> 24 ft, with about ±1.5 ft spread. The ceiling is 25 ft. Which pressure should you choose?
Separate investigation observations:
Chart: 45 psi -> 15 ft, 55 psi -> 19 ft, 65 psi -> 22 ft.
Spread: About ±1 ft at each pressure.
Venue: Ceiling announced at 21 ft.

answer: A single launch might fly near the top of the spread; if that exceeds the ceiling, the 0.25 multiplier wipes out the flight. (4 points).

8. (4 points) What must a team do to launch at its chosen pressure?

Unit: How an air-powered bottle rocket climbs.
Worked-case data: A rocket reaches apogee 1.2 s after launch, and the payload then descends for 11.5 s. What share of time aloft came from the descent?
Separate investigation observations:
40 psi: Peak about 12 ft; time aloft 8.1 s.
55 psi: Peak about 18 ft; time aloft 12.4 s.
70 psi: Peak about 24 ft; time aloft 16.0 s in a 30 ft hall.

answer: Justify that pressure using its practice log data, which the supervisor checks along with the gauge reading. (4 points).

9. (4 points) Suggest two useful additional parameters and explain why.

Unit: The practice log and launch decisions.
Worked-case data: A log has 12 flights recording pressure, height, time, and fin count. Is it complete?
Separate investigation observations:
Log entries: 14 flights recorded.
Columns: Pressure, peak height, time aloft, parachute diameter.
Flights 1-3: Peak height column left blank.

answer: Parachute diameter, because it controls descent rate; rocket mass, because it affects climb and stability. (4 points).

10. (4 points) Name one way a parachute can fail to open and a fix.

Unit: Parachute physics: drag and slow descent.
Worked-case data: A 4.7 g payload uses a 40 cm diameter dome (C_d about 1.5). Estimate its descent speed and the time to fall 7 m.
Separate investigation observations:
30 cm canopy: Opens every time; 7.8 s descent.
60 cm canopy: Opens 6 of 10 times; 14.9 s when open, 3 s when not.
50 cm canopy, looser fold: Opens 10 of 10; 12.6 s descent.

answer: Static or tight folding makes plastic stick; fold loosely, keep lines untwisted, and test the packing method repeatedly. (4 points).

11. (2 points) What caused the corkscrew flight?

Supplied synthetic observations:
Flight 1: Rocket corkscrews and lands near a wall.
Inspection: One fin is twisted about 8° from the body line.
Flight 2: Fin realigned: rocket flies straight up.

A. A blunt nose

B. A misaligned fin

C. Too little pressure

Correct: B

12. (2 points) What is the problem with the team's plan?

Supplied synthetic observations:
Ceiling: 24 ft gym.
Log data: 60 psi: about 20 ft peak, 15 s aloft; 75 psi: about 26 ft peak.
Team plan: Launch at 75 psi for "more height."

A. At 75 psi the rocket is likely to strike the 24 ft ceiling

B. At 75 psi the parachute cannot open

C. Higher pressure is never allowed

Correct: A

13. (2 points) What does the data suggest?

Supplied synthetic observations:
Cup depth 2 cm: Payload left during climb in 4 of 5 flights.
Cup depth 4 cm: Payload stayed attached in 3 of 5 flights.
Cup depth 3 cm: Payload released near apogee in 5 of 5 flights.

A. Deeper cups always work better

B. Cup depth has no effect

C. A 3 cm cup depth balances retention and release

Correct: C

14. (2 points) Define "Clinometer" in the context of Ping Pong Parachute.

answer: Tool that measures the angle of elevation to an object. (2 points).

15. (2 points) Define "Impulse" in the context of Ping Pong Parachute.

answer: Thrust times time; the change in momentum it produces. (2 points).

16. (2 points) Define “Additional parameters” in the context of Ping Pong Parachute.

answer: At least two more chosen by the team. (2 points).

17. (2 points) Define “Nose cup” in the context of Ping Pong Parachute.

answer: Open holder on top of the rocket that carries the payload. (2 points).

18. (2 points) Define “Safety margin” in the context of Ping Pong Parachute.

answer: Planned gap between expected apogee and the ceiling. (2 points).